Midterm Information

1 General Information

The midterm will take place on **Friday, October 23rd** from **2:10pm–4:10pm** in **FE 230**.

Please arrive on time. Do not enter the room until the teaching team has completed setup and you are instructed to enter.

You may bring **one single-sided, 8.5 x 11 inches, handwritten aid sheet**. The sheet must be handwritten and may only contain writing on one side. If these instructions are not followed, the aid sheet will not be permitted during the exam and the situation may be treated as an academic integrity concern.

Calculators are not permitted. All electronic devices, including cellphones and watches, must be turned off and stored in your bag for the duration of the exam.

Please bring a valid form of identification, such as your student card or government-issued ID.

2 Structure and Topics

The midterm will cover all course material up to and including the end of the lecture on **generalized linear models (GLMs)**.

Questions will include **True/False, multiple choice, and short-answer questions**. The exam will emphasize **interpretation, conceptual understanding, and statistical reasoning** rather than lengthy calculations. You will therefore not need a calculator.

The lecture slides contain all of the material required for the midterm. The assignment questions, particularly those involving interpretation and explanation, are also good practice.

Some sample questions are included below to give you a sense of the style and level of the exam. These questions are **not intended to be a complete study guide**. You should make sure that you are comfortable with all of the course material, including topics that do not appear in the practice questions.

3 Some Practice Questions

The questions below are examples of the types of questions you may see on the midterm. They are intended to help you practice the style of reasoning expected on the exam.

The topics on the midterm are **not limited to the examples below**, so please make sure that you review all of the course material.

True or False

If false, briefly explain why.

(a) Increasing the sample size generally increases statistical power.

(b) Adding another predictor to an OLS model can never decrease the training R^2.

(c) In weighted least squares, observations with smaller error variance should generally receive larger weights.

(d) If a result is statistically significant, it must also be practically important.

(e) In Poisson regression with a log link, e^{β_1} has a multiplicative interpretation on the expected count scale.

(f) Increasing the sample size will always fix a misspecified variance assumption in a generalized linear model.

(g) A large F-statistic provides evidence against the joint null hypothesis being tested.

(h) Underdispersion in a Poisson regression model will generally inflate the Type I error rate.

Multiple Choice

There may be more than one correct answer.

(a) Which of the following involve questionable statistical practice?

i. A researcher checks residual plots after fitting a pre-specified regression model.

ii. A researcher tests five different outcomes and reports only the one with $p < 0.05$, without mentioning the other tests.

iii. A researcher explores several subgroups, finds one interesting pattern, and clearly labels it exploratory.

iv. A researcher repeatedly checks significance during data collection and stops when $p < 0.05$, without adjusting for the repeated testing.

(b) Which statements are correct?

i. A very small sample from a highly skewed population may cause the usual t-test to have poor Type I error control.

ii. The central limit theorem can make inference more robust to non-normality as the sample size increases.

iii. Any amount of non-normality makes the t-test invalid.

iv. A permutation test may be useful when the usual parametric assumptions are questionable.

(c) Suppose a residual-versus-fitted plot shows a clear curved pattern but roughly constant spread.

Which statements are correct?

i. The mean structure may be misspecified.

ii. The constant variance assumption is definitely violated.

iii. Adding a nonlinear term may improve the model.

iv. Weighted least squares is necessarily the appropriate fix.

Short Answer

1. Suppose $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ and the variance of ε_i increases as X_i increases.

   (a) Why might weighted least squares be preferred to ordinary least squares?

   (b) Which observations should receive greater weight?

   (c) Does heteroscedasticity necessarily make $\hat{\beta}_1$ biased? Explain.

2. These questions are related to model selection

   (a) Describe how **forward selection** works.

   (b) We want to compare several models for predicting the number of hours patients spend in the hospital after surgery. The response distribution is highly irregular, and we do not want to assume that it follows a particular parametric distribution.

   Which model-selection approach would you recommend?

3. We perform a large number of hypothesis tests in the same study.

   (a) Explain the difference between the family-wise error rate (FWER) and the false discovery rate (FDR).

   (b) Give an example of a situation where controlling the FWER would be preferred over controlling the FDR.

   (c) Give an example of a situation where controlling the FDR would be preferred over controlling the FWER.

4. A researcher studies whether a patient develops a post-surgical infection ($Y_i = 1$ if yes, $Y_i = 0$ if no) using the logistic regression model

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i,$$

   where X_i is the duration of surgery in hours.

   (a) What is p_i in this model?

   (b) Interpret β_1 in the context of this study.

   (c) Suppose $\beta_1 > 0$. What does this tell us about the relationship between surgery duration and the probability of infection?

   (d) Why would ordinary linear regression generally not be appropriate for this response variable?

5. Consider a Poisson regression model for count data. The fitted mean structure appears appropriate, but the observed conditional variance is systematically smaller than the fitted conditional mean.

(a) What does this suggest about the dispersion parameter ϕ?

(b) Explain why this does not necessarily imply that the estimated regression coefficients are biased.

(c) What consequence would you expect for the Type I error rate?